CourseName:CryptographyandNetworkSecurity					
CourseCode :21BTCDSTPE713					
	L	T	P	Category	PE
Contact Hrs./Week	3	0	0	InternalMarks	40
ContactHrs./Sem.	42	0	0	Semester Marks	60
Credits.	3	0	0	ExamHours	3

Courseobjectives:

1. UnderstandOSIsecurityarchitectureandclassicalencryptiontechniques.
2. Acquirefundamentalknowledgeontheconceptsoffinitefieldsandnumber theory.
3. Understandvariousblockcipherandstreamciphermodels.
4. Describetheprinciplesofpublickeycryptosystems,hashfunctionsand digitalsignature.

Prerequisites:Knowledgeoncomputernetworksandengineering mathematicsisrequired.

Units	Teaching Hours
Unit-1:-INTRODUCTION	
Introduction to Computer Security Concepts, Security Attacks, Security, Services and Mechanisms, Symmetric Cipher model, Cryptography & Cryptanalysis, Substitution Techniques: Caesar cipher, Mono alphabetic cipher, Play fair cipher, Hill Cipher, Poly alphabetic ciphers: Vignere cipher, Vernam cipher, Trans position cipher.	9
Unit-2:-FINITE FIELDS AND NUMBER THEORY	
Groups, Rings, Fields-Modular arithmetic - Euclid‟s algorithm- Finite fields- Polynomial Arithmetic – Prime numbers- Fermat‟s and Euler‟s theorem- Testing for primality -The Chinese remainder theorem-Discrete logarithms.	7
Unit -3:-BLOCK CIPHERS & PUBLIC KEY CRYPTOGRAPHY	
Data Encryption Standard- Block cipher principles- block cipher modes of operation-Advanced Encryption Standard (AES)-Triple DES- Blowfish- RC5 algorithm. Public key cryptography: Principles of public key crypto systems-The RSA algorithm- Key management – Diffie Hellman Key exchange-Elliptic curve arithmetic- Elliptic curve cryptography.	9
Unit-4:-HASH FUNCTIONS AND DIGITAL SIGNATURES AND SYSTEM SECURITY	
Message Integrity and Authentication; Cryptographic Hash Functions: MD5, SHA family, Digital Signature and applications – El Gamal. Firewalls for Trusted System: Roles of Firewalls – Firewall related terminology-Types of Firewalls–Firewall designs–SET for E-Commerce Transactions. Intruder–Intrusion detection system–Virus and related threats–Countermeasures–Firewalls design principles– Trusted systems –Practical	9

Implementation of cryptography and security.	
Unit-5:-E-MAIL,IP& WEBSECURITY	
E-mail Security:Security ServicesforE-mail-attacks possiblethrough E-mail –establishingkeysprivacy-authenticationofthesource-MessageIntegrity-Non-repudiation-PrettyGoodPrivacy-S/MIME.IPSecurity:OverviewofIPSec–IP	8
and IPv6-Authentication Header-Encapsulation Security Payload (ESP)-InternetKeyExchange(PhasesofIKE,ISAKMP/IKEEncoding).WebSecurity:SSL/TLSBasicProtocol-computingthekeys-clientauthentication-PKIasdeployedbySSLAttacksfixedinv3-Exportability-Encoding-SecureElectronic Transaction(SET).	
Courseoutcomes: **CO1:-** Identifysecurityobjectives&threatsandenumeratenecessaryservices&mechanismsforeffective countermeasures. **CO2:-** Explainthemathematicalfoundationofcryptographythroughmodulararithmetic,linearalgebra,numbertheory,factorizationanddiscretelogarithm. **CO3:-** AnalyzetheperformanceoftraditionalsymmetrickeycryptographytechniquesandmodernsymmetrickeyciphersslikeDESandAES. **CO4:-**Applypublickeycryptographyand Hashalgorithmsinencryption,dataintegrity,authentication,digitalsignature,andkeyexchange. **CO5:-** Applycryptographytechniquesinvariousnetworksecurityprotocolslike SSL,TLS,PGP,S/MIME, andIPsec andunderstandE-mailSecurity.	
TextBooks: 1. W.Stallings,**CryptographyandNetworkSecurity:PrincipleandPractice**,7thEdition,PearsonEducation,2017. 2. B.A.ForouzanandD.Mukhopadhaya,**CryptographyandNetworkSecurity**,2ndEdition,McGraw-HillEducation,2010.	

ReferenceBooks:

1. CharlieKaufmanandRadiaPerlman,MikeSpeciner,"NetworkSecurity,SecondEdition,Private CommunicationinPublicWorld",PHI2002.
2. C.P.Pfleeger,S.L.Pfleeger,andJ.Margulies,**SecurityinComputing**,5thEdition,Prentice HallIndia,2015.
3. C.Kaufman,R.Perlman,andM.Speciner,**NetworkSecurity:PrivateCommunication inaPublicWorld**,2ndEdition, PrenticeHallIndia,2002.

OnlineResources:

1. https://nptel.ac.in/courses/106/105/106105031/:byDr.D.Mukhopadhyay,IITKharagpur.
2. https://nptel.ac.in/courses/106/105/106105162/:byProf.S.Mukhopadhyay,IITKharagpur.
3. https://nptel.ac.in/courses/106/106/106106221/:byProf.A.Choudhury,IIITBangalore.

NoofAssignment:3

IndustryExpertTalk:2

Mapping withProgramOutcomes(POs)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1						3		3
CO2	3	3	1	1						3		3
CO3	3	3	3	2		2				3		2
CO4	3	3	3	2		2				3		3
CO5	1	3	3	2		2				3		3